Electrical Circuits II
(EE-1201)

Lecture-02

Network Theorems

(Thevenin's Theorem)

Dr. M. N. Alam, EED, NITW

Outlines

©2026 Dr. M. N. Alam

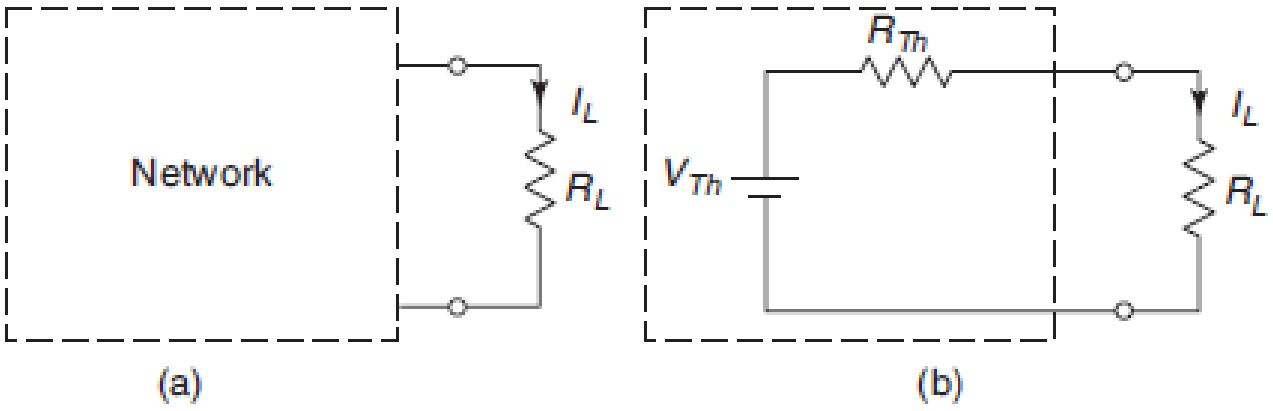
This unit covers:

- Thevenin's theorem
- Numerical examples
- Homework

Thevenin's Theorem

©2026 Dr. M. N. Alam

- It states that *"any two terminals of a network can be replaced by an equivalent voltage source and an equivalent series resistance. The voltage source is the voltage across the two terminals with load, if any, removed. The series resistance is the resistance of the network measured between two terminals with load removed and constant voltage source being replaced by its internal resistance (or if it is not given with zero resistance, i.e., short circuit) and constant current source replaced by infinite resistance, i.e., open circuit."*

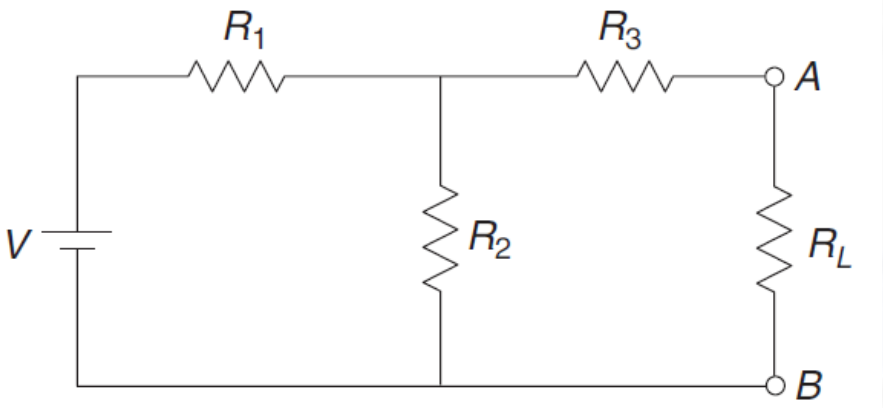


Network illustrating Thevenin's theorem

Thevenin's Theorem

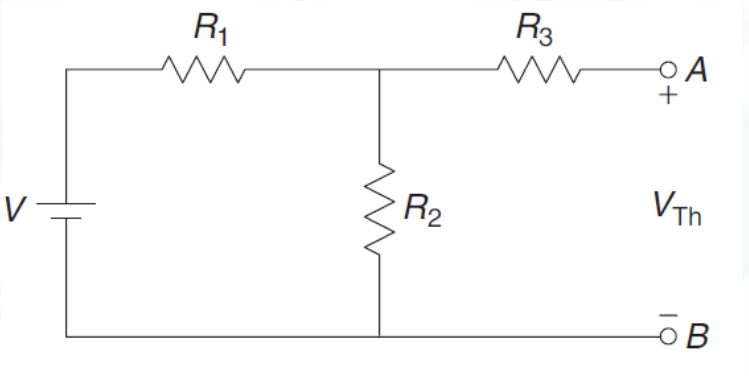
©2026 Dr. M. N. Alam

Explanation: Consider a simple network as shown in Fig.



Step 1: Thevenin's voltage V_{Th} calculation,

For finding load current through $\mathrm{R_L}$, first remove the load resistor $\mathrm{R_L}$ from the network and calculate open circuit voltage V_{Th} across points *A* and *B* as shown in Fig.



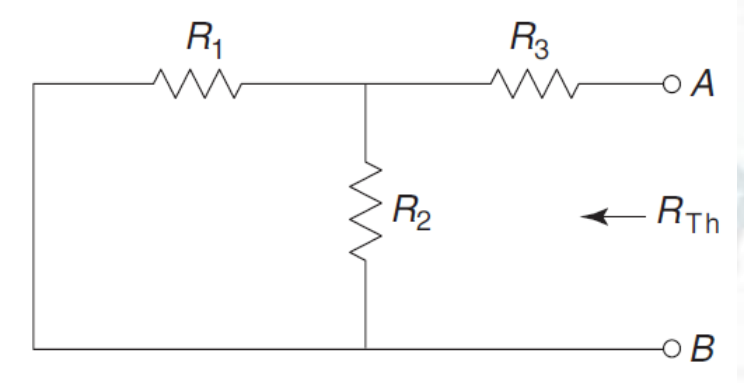
$$V_{Th} = \frac{R_2}{R_1 + R_2} V$$

Thevenin's Theorem

©2026 Dr. M. N. Alam

Step 2: Thevenin's resistance R_{Th} calculation,

For finding series resistance R_{Th}, replace the voltage source by a short circuit and calculate resistance between points A and B as shown in Fig.

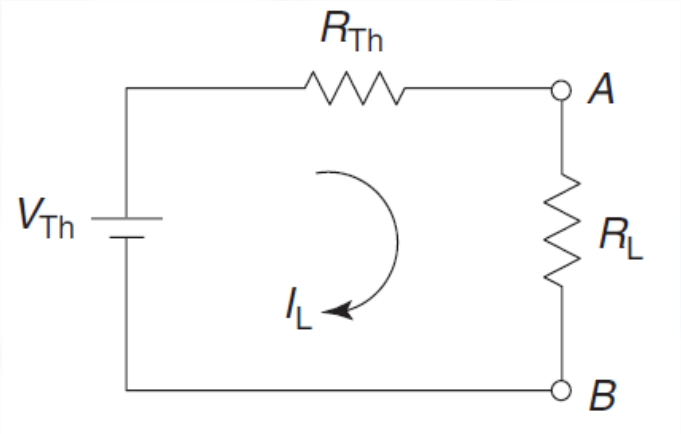


$$R_{Th} = R_3 + \frac{R_1 R_2}{R_1 + R_2}$$

Step 3: Thevenin's equivalent network

Now, load current through $\mathrm{R_L}$ can be expressed as

$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$



Thevenin's equivalent network

Thevenin's Theorem

©2026 Dr. M. N. Alam

Network contains both independent and dependent sources

- Thevenin's resistance R_{Th} is calculated as

$$R_{Th} = \frac{V_{Th}}{I_N}$$

where I_N is the short-circuit current which would flow in a short circuit placed across the terminals A and B. Dependent sources are active at all times. They have zero values only when the control voltage or current is zero.

- R_{Th} may be negative in some cases which indicates negative resistance region of the device, i.e., as voltage increases, current decreases in the region and vice-versa.

Thevenin's Theorem

©2026 Dr. M. N. Alam

Network contains only dependent sources

- If the network contains only dependent sources then

$$V_{Th} = 0$$
$$I_N = 0$$

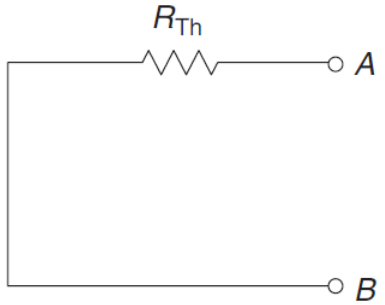
- For finding R_{Th} in such a network, a known voltage V is applied across the terminals A and B and current is calculated through the path AB

$$R_{Th} = \frac{V}{I}$$

- ***Alternatively***, a known current source I is connected across the terminals A and B and voltage is calculated across the terminals A and B

$$R_{Th} = \frac{V}{I}$$

Thevenin's equivalent network for such a network is shown in the Fig



Thevenin's Theorem

©2026 Dr. M. N. Alam

Steps to be Followed in Thevenin's Theorem

1. Remove the load resistance R_L
2. Find the open circuit voltage V_{Th} across points A and B
3. Find the resistance R_{Th} as seen from points A and B
4. Replace the network by a voltage source V_{Th} in series with resistance R_{Th}
5. Find the current through R_L using Ohm's law

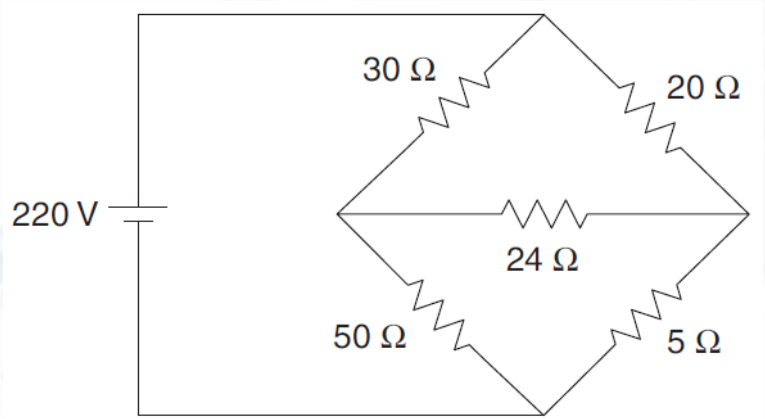
$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

Thevenin's Theorem

©2026 Dr. M. N. Alam

Numerical Example-6:

Determine the current through the 24 Ω resistor using Thevenin's theorem.

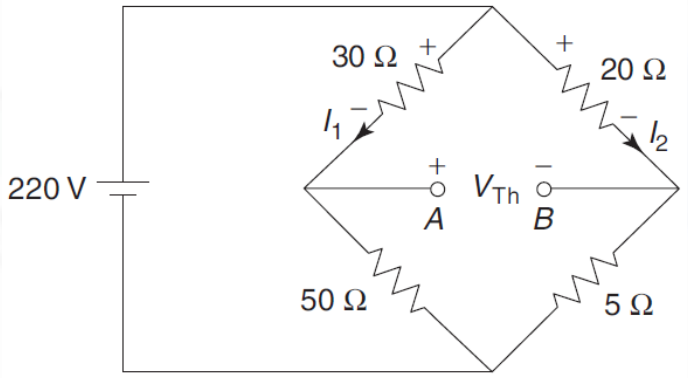


Solution:

Step I: Calculation of V_{Th},

$$I_1 = \frac{220}{30+50} = 2.75\ A$$

$$I_2 = \frac{220}{20+5} = 8.8\ A$$



Thevenin's Theorem

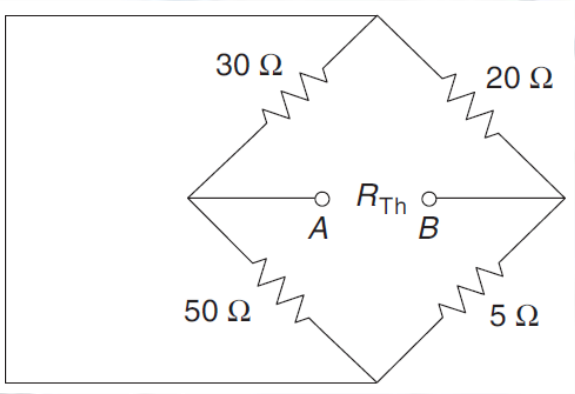
©2026 Dr. M. N. Alam

- Writing voltage equation,

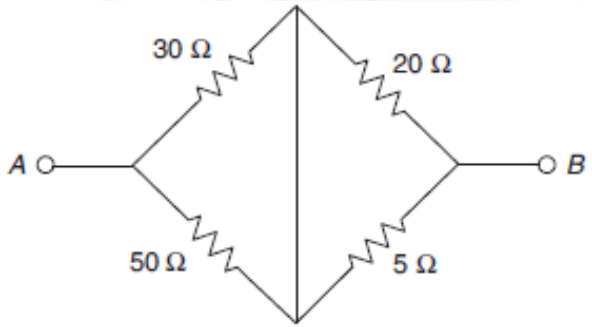
$$V_{Th} + 30I_1 - 20I_2 = 0$$

$$V_{Th} = 20I_2 - 30I_1 = 20 \times 8.8 - 30 \times 2.75 = 93.5\ V$$

Step II: Calculation of R_{Th},



- Redrawing the circuit,

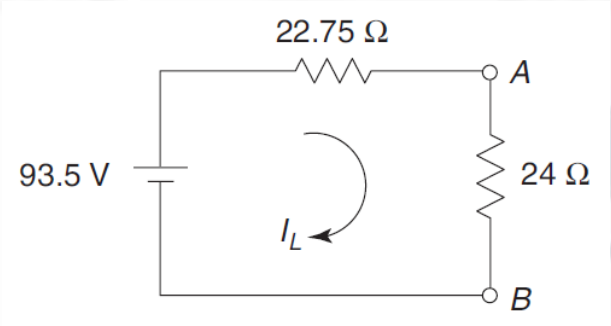


$$R_{Th} = (30||50) + (20||5)$$
$$= 22.75\ \Omega$$

Thevenin's Theorem

©2026 Dr. M. N. Alam

Step III: Calculation of I_L,



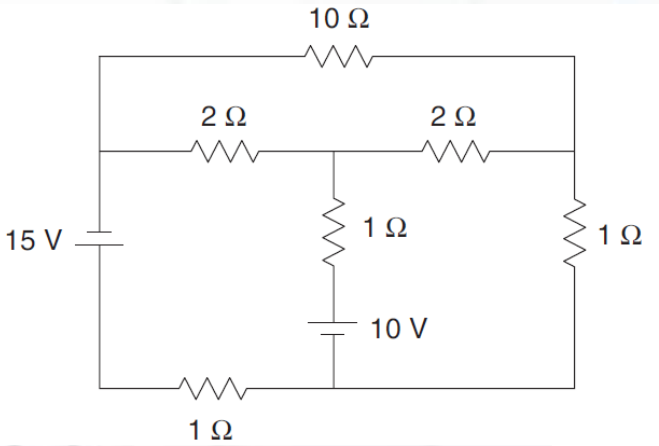
$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{93.5}{22.75 + 24} = 2\ A$$

Thevenin's Theorem

©2026 Dr. M. N. Alam

Numerical Example-7:

Determine the current through the 10 Ω resistor using Thevenin's theorem.



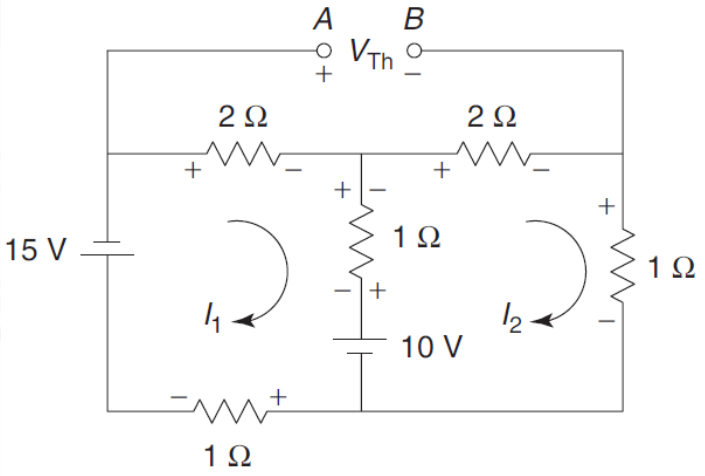
Solution:

Step I: Calculation of V_{Th},

- Applying KVL to Mesh 1

$$-2I_1 - 1(I_1 - I_2) - 10 - 1I_1 - 15 = 0$$

$$-4I_1 + I_2 = 25 \ldots . (i)$$



Thevenin's Theorem

©2026 Dr. M. N. Alam

- Applying KVL to Mesh 2

$$-2I_2 - 1I_2 + 10 - 1(I_2 - I_1) = 0$$

$$I_1 - 4I_2 = 10 \ldots . (ii)$$

- Solving Eqs (i) and (ii),

$$I_1 = -6\ A$$

$$I_2 = 1\ A$$

- Writing voltage equation,

$$-V_{Th} + 2I_2 + 2I_1 = 0$$

$$V_{Th} = 2I_2 + 2I_1 = 2 \times (1) + 2 \times (-6) = -10\ V$$

$$V_{Th} = 10\ \text{V (terminal B is positive)}$$

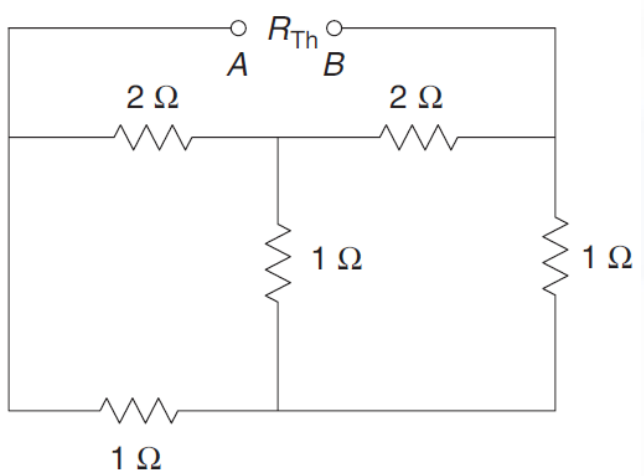
Step II: Calculation of R_{Th},

- Converting the star network formed by resistors of 2 Ω, 2 Ω, and 1 Ω into an equivalent delta network

$$R_1 = 2 + 2 + \frac{2 \times 2}{1} = 8\ \Omega$$

$$R_2 = 2 + 1 + \frac{2 \times 1}{2} = 4\ \Omega$$

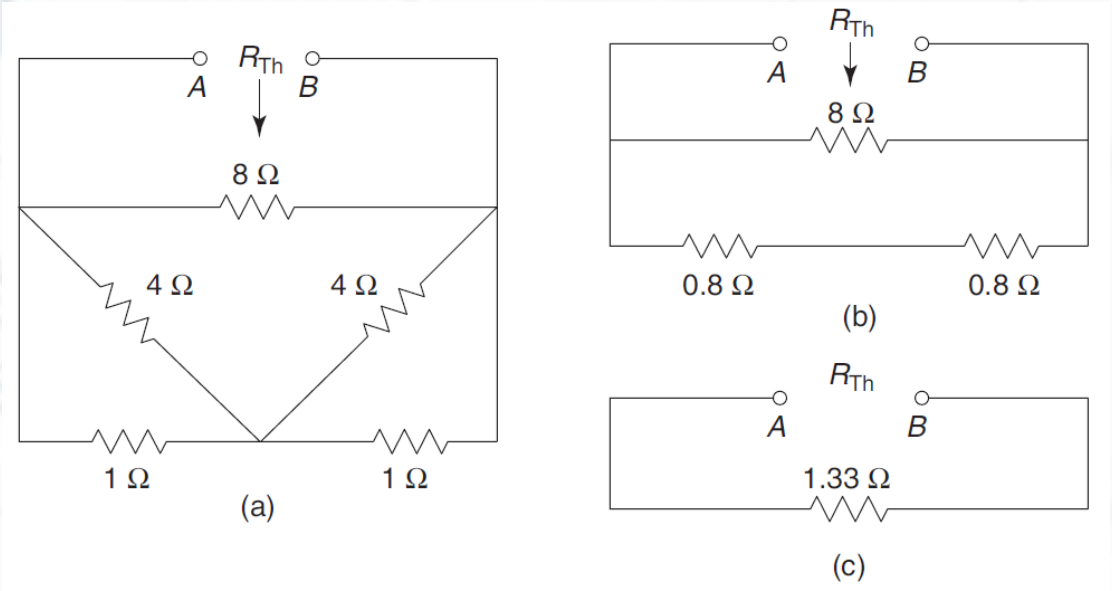
$$R_3 = 2 + 1 + \frac{2 \times 1}{2} = 4\ \Omega$$



Thevenin's Theorem

©2026 Dr. M. N. Alam

- Simplifying the network,

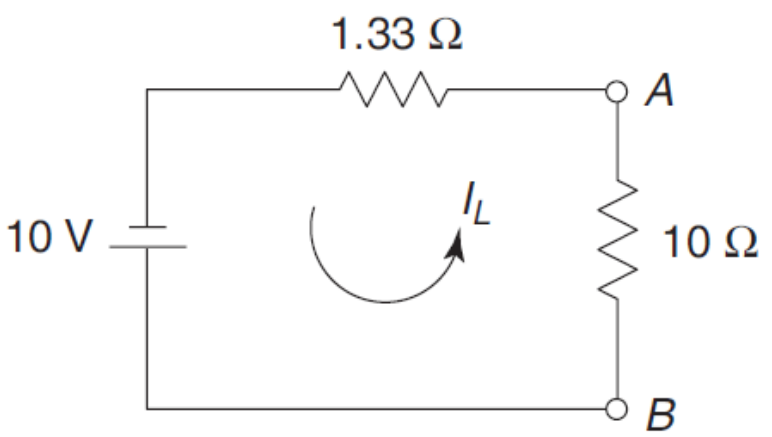


$$R_{Th} = 1.33\ \Omega$$

Thevenin's Theorem

©2026 Dr. M. N. Alam

Step III: Calculation of I_L,



$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{10}{1.33 + 10} = 0.88\ A\ (\uparrow)$$